

The Chipathon 2026

Proposal Review: Group 1 A1-A25

IEEE Solid-State Circuits Society
Technical Committee on the Open Source Ecosystem (TC-OSE)

12 June 2026



[A10] CryptoAccel

ASCON AEAD128 based Cryptographic Hardware Accelerator with AXI Interface
as a plug and play hardware co-processor for various applications.

Team members

Discord name	Affiliation	Experience	Role
lakvlsi_90908	IIT, Bombay	Ph.D. Research Scholar	Team Lead / Presenter
zsteresis	MIT, Manipal / STMicroelectronics	Undergraduate (III)	Team Member
tarun_rs05	IIIT, Bangalore	Undergraduate (II)	Team Member
harshi070852	PESIT, Bangalore	Undergraduate (IV)	Team Member

PROJECT INFORMATION

GOAL:

- ❑ Design and implement a lightweight ASCON-AEAD128a cryptographic hardware accelerator with an AXI-Lite interface using GF180MCU technology node
- ❑ Enable seamless integration of the accelerator with a wide range of processors and System-on-Chip (SoC) platforms that support the AXI protocol.

DESIGN:

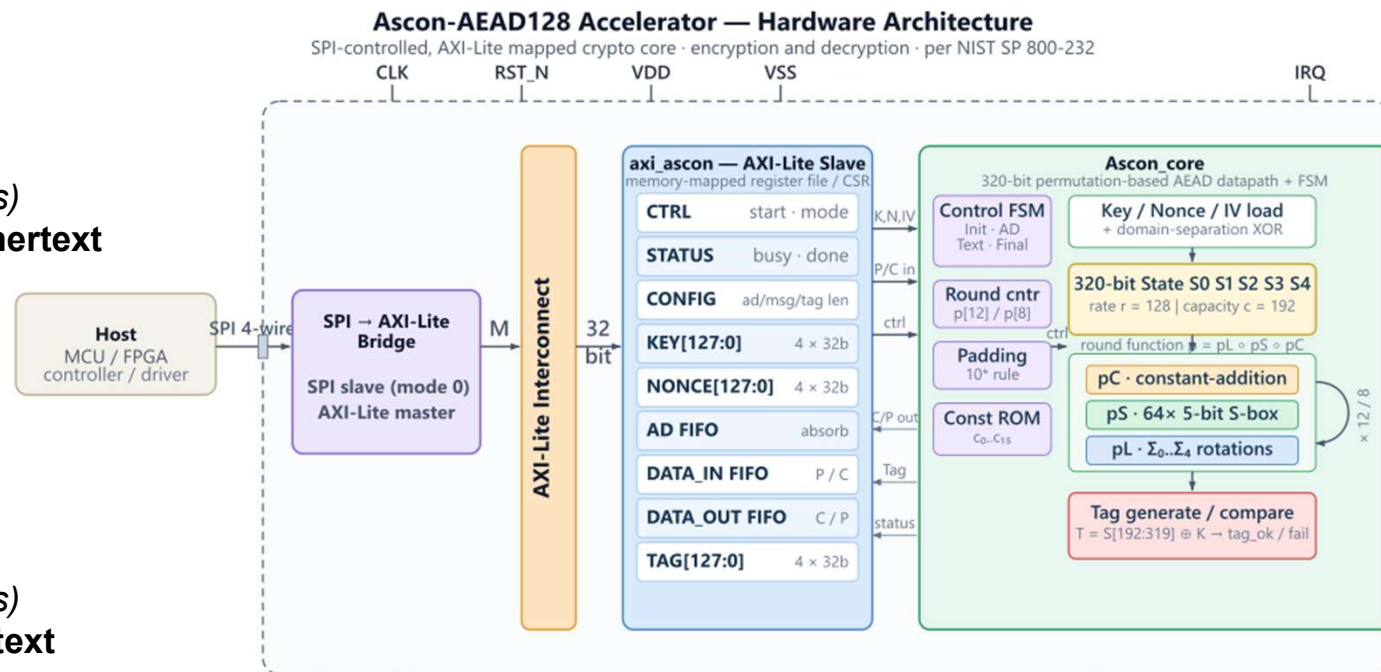
Encryption:

1. Configure and key the engine (*SPI writes*)
2. Start (*SPI write*)
3. Feed the associated data (*SPI writes*)
4. Stream plaintext and collect ciphertext (*SPI writes + reads, interleaved*)
5. Finish and read the tag (*SPI reads*)

Decryption:

1. Configure and key the engine (*SPI writes*)
2. Start (*SPI write*)
3. Feed the associated data (*SPI writes*)
4. Stream ciphertext and collect plaintext (*SPI writes + reads*)
5. Verify (*SPI read*)

ARCHITECTURE



APPLICATIONS:

- ❑ Internet of Things (IoT) devices: authenticated encryption and data integrity protection for sensor nodes, edge devices, and embedded controllers operating under strict area and power constraints.
- ❑ Serve as a hardware Root of Trust for secure boot applications by verifying the authenticity and integrity of firmware during system startup.

SPECIFICATIONS

Technology	GF180MCU
Supply Voltage	3.3V
Frequency	50MHz
AREA	<0.8 mm ²
PIN COUNT	~ 15 PINS

REFERENCES

- [1] M. Sönmez Turan, K. McKay, J. Kang, J. Kelsey, and D. Chang, "Ascon-Based Lightweight Cryptography Standards for Constrained Devices: Authenticated Encryption, Hash, and Extendable Output Functions," NIST Special Publication 800-232, Aug. 2025.
- [2] K. -D. Nguyen, T. -K. Dang, B. Kieu-Do-Nguyen, C. -K. Pham and T. -T. Hoang, "Live Demonstration: ASIC Implementation of ASCON Lightweight Cryptography for IoT Applications," 2025 IEEE International Symposium on Circuits and Systems (ISCAS), London, United Kingdom, 2025, pp. 1-1, doi: 10.1109/ISCAS56072.2025.11044102.
- [3] J. Kaur, A. C. Canto, M. M. Kermani, and R. Azarderakhsh, "A Comprehensive Survey on the Implementations, Attacks, and Countermeasures of the Current NIST Lightweight Cryptography Standard," Electronics, vol. 13, no. 22, Art. no. 4550, 2024.
- [4] OpenTitan Project, "EarlGrey and Darjeeling Architectures,"
[Online]. Available: <https://opentitan.org/book/doc/productarchitecture.html>
- [5] Khan, S., Inayat, K., Muslim, F.B. et al. Securing the IoT ecosystem: ASIC-based hardware realization of Ascon lightweight cipher. Int. J. Inf. Secur. 23, 3653–3664 (2024). <https://doi.org/10.1007/s10207-024-00904-1>